

CSSE3113 – Introduction to Software Engineering

SOFTWARE QUALITY ASSURANCE

Software quality assurance vs. Software quality control

- Quality Control (QC) is a set of activities designed to evaluate the quality of developed or manufactured product(IEEE,1991)
- Activities whose main objective is the withholding of any product that does not qualify.
- Quality Assurance (QA) is meant to minimize the costs of quality by introducing a variety of activities throughout the development process and maintenance process in order to prevent the causes of errors, detects them, and corrects them in the early stages of the development.

Definition: Software Quality Engineering

Software Quality Engineering:

“ The function of software quality is to assure that quality is built into the software by performing analysis, investigations on the requirements, design, code and verification processes and results to assure that reliability, maintainability, and other quality factors are met.”

or

“The application of a continuous, systematic, disciplined, quantifiable approach to the development and maintenance of quality throughout the whole life cycle of software products and systems; that is, the application of quality engineering to software”

Software errors, faults and failures

Software errors:

Sections of the code that are partially or totally incorrect as a result of a grammatical, logical or other mistake made by a systems analyst, a programmer, or another member of the software development team.

Software faults

are software errors that cause the incorrect functioning of the software in general or in specific application.

Software faults become software failures only when they are “activated”, that is, when a user tries to apply the specific software section that is faulty. Thus, the root of any software failure is a Software error.

Do all software faults end with software failures?

Software Defect Origins

- As software errors are the cause of poor software quality, it is important to investigate the causes of these errors in order to prevent them.
- The causes of software errors can be further classified as follows according to the stages of the software development process in which they occur.
 - Faulty definition of requirements
 - Client-side mistakes
 - Absence of vital requirement
 - Incomplete definition of Requirement
 - Inclusion of unnecessary requirements

Software Defect Origins

Client–developer communication failures

Misunderstanding of the

- Client's instructions as stated in the requirement document
- Client's requirements changes presented to the developer
- Client's responses to the design problems presented by the developer.
- Lack of attention to client messages referring to requirements changes

Deviations from software requirements

- Developers may deliberately deviate from the documented requirements
- The developer reuses software modules taken from an earlier project without sufficient analysis of the changes
- Due to time or budget pressures, the developer decides to omit part of the required functions

Software Defect Origins

Logical design errors

- Algorithm mistakes
- Non-compliance with documentation and coding standard

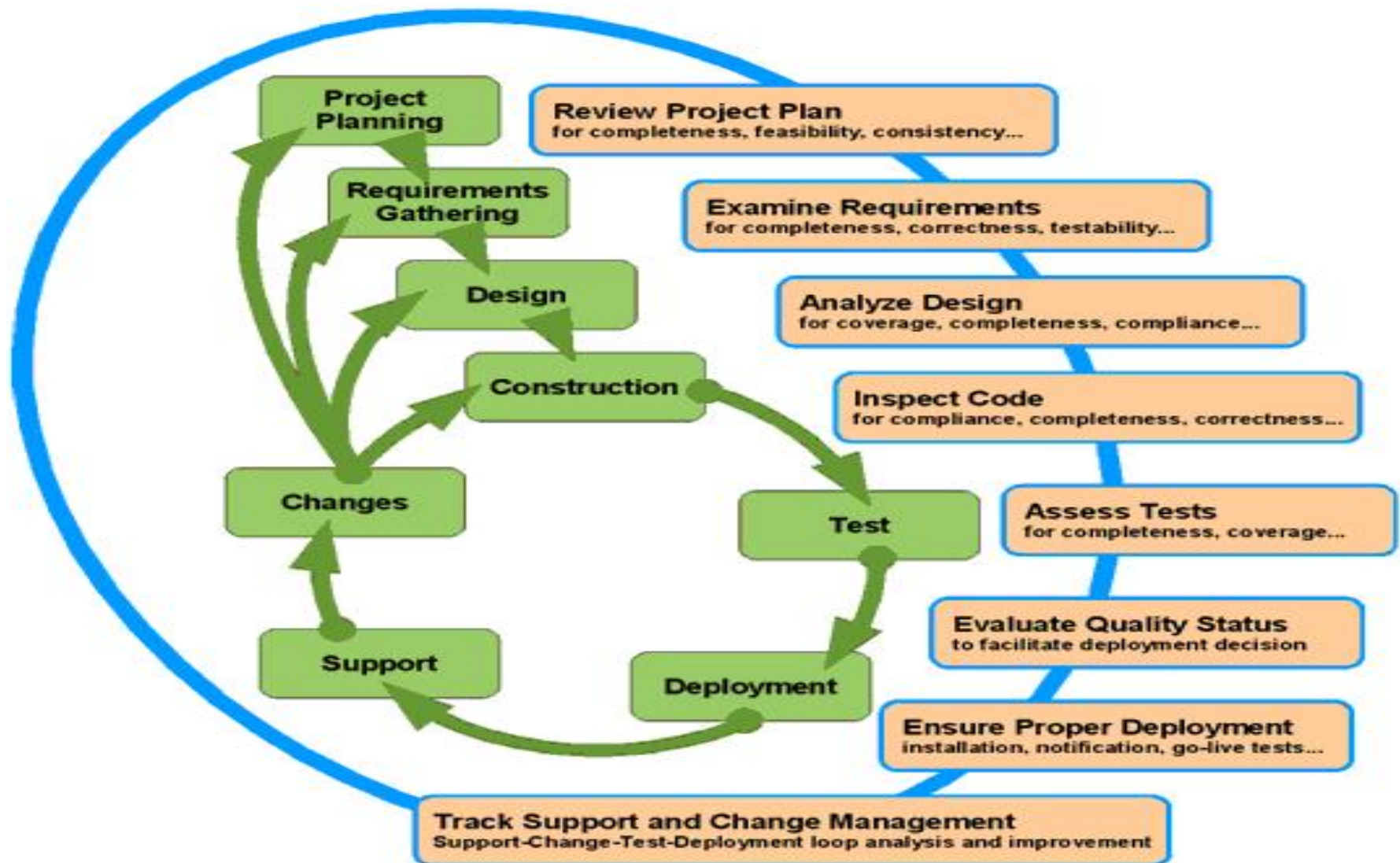
Shortcomings of testing process

- Incomplete test plans leave untreated portions of the software
- Incomplete correction of detected errors due to negligence or time pressures.

Documentation Errors

- Errors in design document
 - Omission of software functions.
- Errors in user manual
 - Errors in the explanations and instructions given to users

SQA – Software Life Cycle



Quality Standards

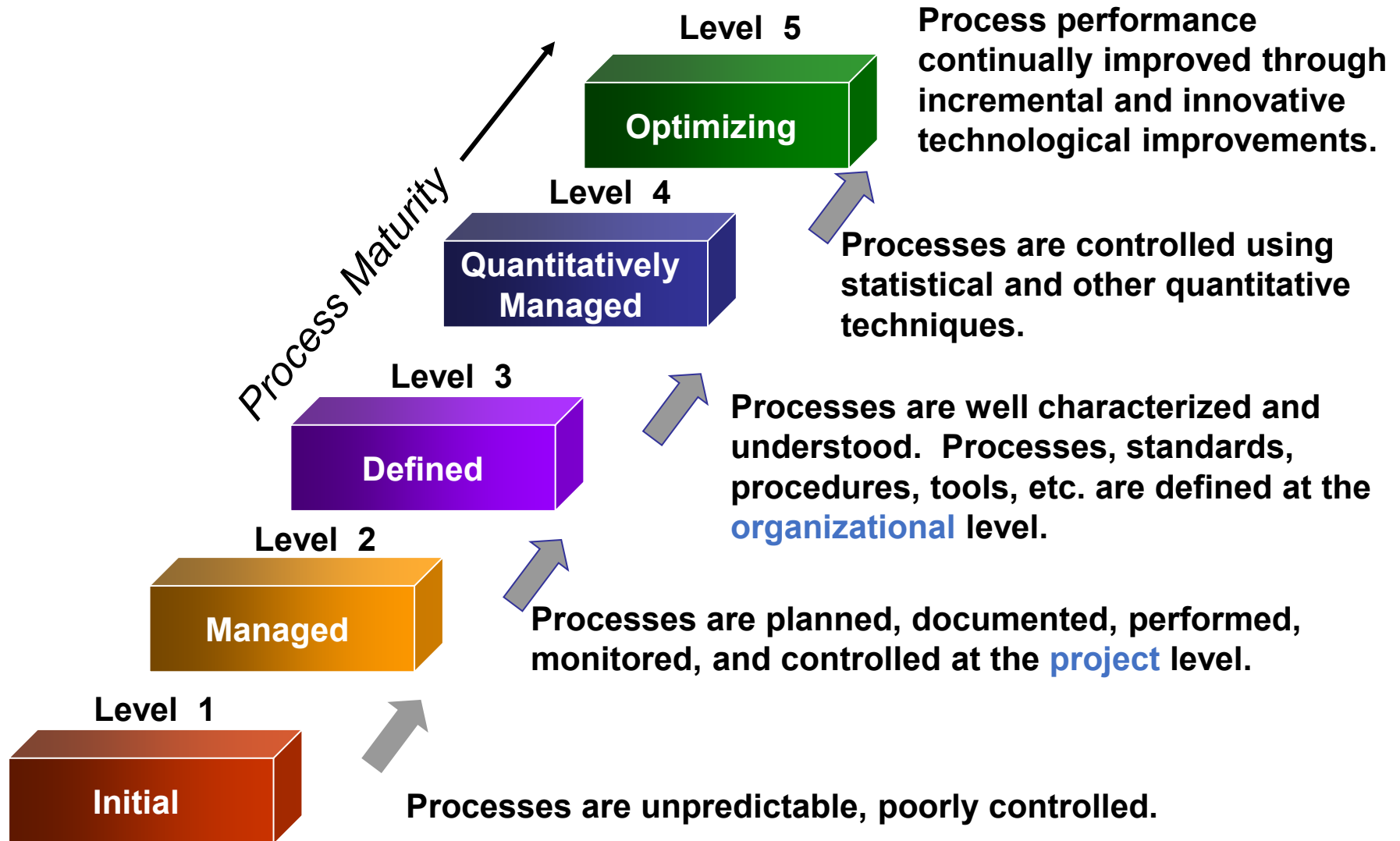
Quality standards for software engineering:

These are standard written down, that must be followed by software engineers and engineering firms to achieve quality in their products. Below are some major standards in the software engineering industry:

1. **Capability Maturity Model (CMM):**

written by the software engineering institute, this model describes good practices for planning, engineering and managing software development. This practice allows software engineering organizations to create quality software while minimizing budget and time factors. CMM allows software engineers to judge and compare its processes to stated practices in the industry. Thus improving software processes and quality.

CMMI Staged Representation - 5 Maturity Levels



Quality Standards

2. ISO 9000 family: These are standards for quality management and quality assurance for any business. It applies to process of creating, and controlling of products and services an organization provides, by giving an organized course for activities to be carried out to ensure that customer's requirements are made (ISOQAR, 2010). This includes the:

- ISO 9000
- ISO 9001
- ISO 9126

Quality Standards

3. IEEE STD 1061-1998 Standard:

This standard was created as a way to establish quality requirements, and provides a well defined metrics for identifying, implementing, analyzing and validating the processes of developing a quality software product. The standard provides metrics for quality for the entire software engineering life cycle (IEEE Std 1061, 1998).

Software Quality Assurance and Reviews

- Software development Process → Different documents (SRS+ DDS)
- The system analysts & development team members who prepared the document will check it repeatedly→ in order to detect any possible error that might have entered.
- Others – such as peers, experts, and customer's representatives—reviewing the product and detecting the errors unnoticed by the development team.

Review Methodologies

Several methodologies can be implemented when reviewing documents.

- Formal Reviews
- Peer Reviews (inspections and walkthroughs)
- Expert opinions

Review Methods –Formal reviews

- Formal reviews, variously called ...“formal technical reviews (FTR)”
- Necessary for approval of the product.
- Formal reviews may be conducted at any development milestone→ completion of document, whether that document is a requirement specification , Design document or an installation plan.

Peer Reviews: Inspections / Walkthroughs

- Walkthroughs and inspection differ in formality
 - Inspection the more formal of the two
- Inspections emphasize the objective of corrective action
- Walkthroughs limited to comments on document reviewed.
- Inspections are considered to contribute more to SQA.

The Peer Review Session

In inspection

- Presenter reads a section of the document.
- As the session progresses, Participants either deliver their comments to the document or address their reactions to the comments.
- Restrict discussion to identification of errors.
- Presenter in walkthrough provides an overview
- Walkthrough Scribe records each error (location, description, type and character (incorrect, missing parts, extra parts etc)).
- Inspection scribe will add estimated severity of each defect and the formulation of preventive / corrective actions.

Comparison of review methodologies

Properties	Formal design reviews	Inspections	Walkthroughs
■ Main direct objectives	(1) Detect errors (2) Identify new risks (3) Approve the design document	(1) Detect errors (2) Identify deviations from standards	Detect errors
■ Main indirect objectives	Knowledge exchange	(1) Knowledge exchange (2) Support corrective actions	Knowledge exchange
■ Review leader	Chief software engineer or senior staff member	Trained moderator (peer)	Coordinator (peer, the project leader on occasion)
■ Participants	Top-level staff and customer representatives	Peers	Peers
■ Project leader participation	Yes	Yes	Yes; usually as the review's initiator
■ Specialized professionals in the team	—	(1) Designer (2) Coder or implementer (3) Tester	(1) Standards enforcer (2) Maintenance expert (3) User representative

Comparison of review methodologies

Properties	Formal design reviews	Inspections	Walkthroughs
Process of review:			
■ Overview meeting	No	Yes	Yes
■ Participants' preparations	Yes – thorough	Yes – thorough	Yes – brief
■ Review session	Yes	Yes	Yes
■ Follow-up of corrections	Yes	Yes	No
Infrastructure:			
■ Formal training of participants	No	Yes	No
■ Use of checklists	No	Yes	No
Error-related data collection	Not formally required	Formally required	Not formally required
Review documentation	Formal design review report	(1) Inspection session findings report (2) Inspection session summary report	Walkthrough session findings report

Expert Opinions

- Expert opinions support quality assessment efforts by introducing additional external capabilities into the organization's in-house development process
- Turning to outside experts may be particularly useful in the following situations:
 - Insufficient in-house professional capabilities in a given area
 - In small organizations: difficult to find enough suitable candidates to participate in the design review teams
 - In small organizations: extreme work pressures, an outside expert's opinion can replace an inspection process
 - Temporary inaccessibility of in-house professional
 - In cases of major disagreement among the organization's senior professional, an outside expert may support a decision